Amir Kashani

Senior Systems & Connectivity Architect — Digital Car Key (CCC 3.0 / CarKey) & High-Scale Mobile Platforms

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PROFILE

Senior Systems & Connectivity Architect with 10+ years architecting iOS, Android, and shared C++ systems, specializing in CCC Digital Key 3.0, Apple CarKeyKit, UWB, BLE, NFC, and Secure Element (SE) hardware attestation. Proven track record leading mobile platform architecture for 1M+ users and 400K+ subscribers (+30% retention and load performance), reducing video startup from 4–5s to under 1 second, and delivering multi-platform products from 0-to-1. Creator of Recordium, featured globally by Apple in 110 countries and ranked Top 10 Business App worldwide.

CORE STACK

Automotive & Hardware Security: CCC Digital Key 3.0 · Apple CarKeyKit · CarKeySession · PassKit · Secure Element (SE) · UWB Distance Ranging · BLE Proximity & GATT · NFC APDU · ECU/OEM Gateway integration · Cryptographic key sharing

Systems Architecture & Languages: Swift · Modern C++ · Objective-C · Kotlin / JNI Interop · Python · SwiftUI · UIKit · Combine · Async/Await · GCD · Clean Architecture · Domain-Driven Design (DDD) · Modular Frameworks · IPC

Media & Infrastructure: Distributed VOD Architecture · AVPlayer ABR Streaming · Multi-CDN Edge Failover · Offline DRM & Encrypted Caching · SQLite · Core Data · CI/CD Automation · Release Regression Testing

AI & Developer Velocity: Multimodal Video Pipelines (Qwen) · Automated LLM Crash Triage · AST Codebase Knowledge Graphs · Whisper & pyannote Diarization · Local Motion Detection (MacSpyCam)

EXPERIENCE

Senior Connected Mobility Systems Engineer · Quartett Mobile

MAR 2022 — PRESENT

Connected Vehicle & German Automotive OEM Programs · Munich, Germany

- Architected and delivered end-to-end digital vehicle access provisioning workflows utilizing **Apple CarKeyKit**, **CarKeySession**, and **PassKit**, managing cryptographic owner pairing, Apple Wallet key provisioning, and vehicle credential synchronization.
- Implemented **CCC Digital Key 3.0** cryptographic security workflows, orchestrating **Secure Element (SE)** applet attestation, certificate exchange, and **NFC APDU** contact transactions (tap-to-unlock and engine start) to enforce zero-trust access control against relay attacks.
- Integrated multi-transport passive entry and vehicle localization across **BLE** proximity detection and **UWB** distance ranging in close collaboration with vehicle **ECU/OEM Gateway**, embedded firmware, and cloud engineering teams.
- Decoupled and co-developed a cross-platform **Shared C++ Core** state machine for **BLE** transport and cryptographic protocol orchestration, eliminating cross-platform protocol divergence and duplicate maintenance overhead across iOS and Android codebases.
- Standardized cross-organizational API contracts and automated mock-based contract testing across distributed mobile, embedded firmware, and vehicle **ECU/OEM Gateway** teams, establishing strict regression quality gates for multi-tier automotive deliveries.

Lead Mobile Systems Architect · Sabaldea Group

MAR 2020 — MAR 2022

Filimo · Televika · Zabia · High-Scale Video Streaming Platforms

- Spearheaded platform-wide architecture modernization to modular **Clean Architecture** and Domain-Driven Design (DDD) for high-concurrency VOD streaming applications (Filimo, Televika, Zabia), scaling throughput to **1M+ active users** and **400K+ subscribers** while boosting retention and load performance by **+30%**.
- Optimized playback pipeline performance, cutting video startup latency from **4–5 seconds to under 1 second** through custom **AVPlayer ABR** (Adaptive Bitrate) streaming heuristics, dynamic multi-CDN failover, and proactive buffer pre-warming.
- Architected secure **Offline DRM** playback and encrypted media caching subsystems with hardware-backed key persistence and SQLite storage, enabling compliant offline viewing across premium licensed catalogs.
- Engineered automated release quality gates, mock network injection suites, and end-to-end regression pipelines, shrinking release regression cycles by **40%** while accelerating deployment velocity.
- Coached and guided cross-functional mobile engineering squads across multiple streaming products, establishing strict API contract governance, watch-history synchronization protocols, and cross-team interface boundaries with backend and data platforms.

Co-Founder & CTO · BeautyTime

NOV 2017 — MAR 2020

Marketplace & B2B SaaS · Multi-platform product and backend infrastructure

- Architected and delivered the complete 0-to-1 omnichannel platform across **3 platforms (Web, iOS, and Android) in 4 months** from inception to initial commercial launch.
- Engineered scalable backend microservices, REST APIs, and database architecture to support **9,000+ peak daily active users (DAU)** with continuous operational availability.
- Engineered real-time B2B SaaS scheduling and B2C geo-discovery marketplace workflows with dynamic booking services, reducing booking abandonment and improving salon operational efficiency.
- Directed a **6-person cross-functional team** spanning engineering, product, marketing, and operations under tight capital constraints, establishing agile sprint cadences and delivery standards.

Senior iOS Engineer · Sheypoor

AUG 2015 — OCT 2017

Large classifieds marketplace (350+ employees) · iOS and platform APIs

- Architected and shipped Sheypoor iOS v2 in **3 months** ahead of an institutional venture funding round for an organization of **350+ employees**.
- Partnered with Android, web, and backend teams on v3 REST API contract specifications and client caching layers, driving a **+20% platform engagement** increase.
- Earned rapid promotion to Senior Engineer within **12 months** based on architectural contributions, technical leadership, and cross-functional delivery.

Mobile Software Engineer · ivysoft

JUN 2013 — JAN 2015

Commercial iOS products across finance, payments, and media

- Optimized application cold-launch performance and memory utilization across flagship commercial iOS products (Rahavard365, Digipay, MusicMa) using Instruments profiling and view hierarchy optimization.
- Engineered telemetry-driven conversion funnels and user analytics, facilitating data-backed transitions from paid downloads to freemium monetization models across production apps.

SELECTED ENGINEERING PROJECTS & RECOGNITION

Recordium — iOS Audio Engine

Global Apple Recognition

Architected and shipped Recordium, an independent iOS audio recording engine featuring real-time waveform visualization and in-audio annotations. **Featured by Apple on the App Store home page across 110 countries**, ranked among the **Top 10 Business Apps worldwide**, and covered in *The Guardian*, *The Telegraph*, and *The Next Web*.

- **Multimodal Video Classification:** Engineered a Qwen vision-language pipeline for automated keyframe extraction, scene semantics, and taxonomy tagging, cutting manual video labeling effort by over 80%.
- **AI Crash Triage Pipeline:** Developed an LLM crash triage pipeline mapping stack traces to deterministic AST codebase knowledge graphs and git blame history, cutting incident triage turnaround from 30 minutes to under 5 minutes.
- **Transcription & Diarization:** Implemented Whisper and pyannote workflows delivering structured, speaker-labeled JSON outputs for multi-speaker media.
- **Codebase Knowledge Graphs:** Built deterministic AST parsing and dependency graphs for navigating and refactoring unfamiliar enterprise repositories.
- **MacSpyCam:** Authored a privacy-focused macOS edge security utility with local motion detection and zero cloud telemetry.

EDUCATION, ACADEMIC WORK & LANGUAGES

Education

M.Sc. in Information Technology (E-Commerce) — Iran University of Science and Technology (2015–2018)

Thesis: Assessing the business value of big-data analysis in firms

B.Sc. in Software Engineering — Shahid Beheshti University (2008–2013)

Academic & Community Work

- **E-Commerce Laboratory (IUST):** Research, seminars, and presentations on big data analytics, text mining, and social-network analysis.
- **SM-Art Robotic Group:** Autonomous control algorithms for RoboCup Small Size League.
- **Chanteh Group:** Designed and taught a native Swift software-development curriculum.

Languages & Honors

Languages: English (Professional working proficiency) · Persian (Native) | **Honors:** Harkat National Festival — Selected Scientific Association Award (2011)